# Predicting Bridge Performance under Earthquakes with Supervised Learning

UE24CS352A – Machine Learning · Mini-Project Write-up

## 1. Problem Statement

Seismic bridge design has moved from capacity-based to performance-based design, yet how well an existing bridge will behave in a future earthquake remains hard to predict. We build a **binary classifier** that predicts whether a bridge is **damaged (1)** or **undamaged (0)** in an earthquake from three inputs: the **age of the bridge**, the **earthquake magnitude** and the **distance from the bridge to the epicenter**. Logistic Regression, Quadratic Discriminant Analysis (QDA) and K-Nearest Neighbours (KNN) are trained and compared.

## 2. Dataset

- **Source:** positive examples (damaged bridges) were collected from U.S. government reports on historic earthquakes; negative examples come from the USGS earthquake catalog. Assumption: if an earthquake occurred near a bridge and no damage is recorded, the record is a negative example.
- **Size:** 249 bridge–earthquake records, of which **21 are damaged and 228 undamaged** (about 1:11, a strongly imbalanced problem). Split used by the code: **199 training** (18 damaged) and **50 test** (3 damaged) records; a separate 149/50 split is used to choose K for KNN.
- **Features** (all continuous): `x2` = bridge age (years), `x4` = earthquake magnitude, `x5` = distance to epicenter (miles). Label `y1` = damaged or not. Material and bridge type are recorded but not used.

## 3. Approach

- **Class imbalance:** the damaged (positive) training examples are up-sampled by **bootstrap resampling with replacement** so that #positive / #negative takes the values 0.2, 0.4, 0.6, 0.8 and 1.0. Because resampling is random, every model is trained **100 times per ratio** and the accuracies are averaged.
- **Logistic Regression** – maximises the log-likelihood of a linear decision boundary. **QDA** – a generative model that fits one Gaussian (own mean and covariance) per class and picks the class with the higher likelihood, giving a quadratic boundary. **KNN** – majority vote among the K nearest training points; K was chosen from 1–10 by validation accuracy (K = 2).
- **Evaluation:** training accuracy, test accuracy, and test accuracy separately on the damaged and undamaged classes, since overall accuracy alone is misleading on imbalanced data.

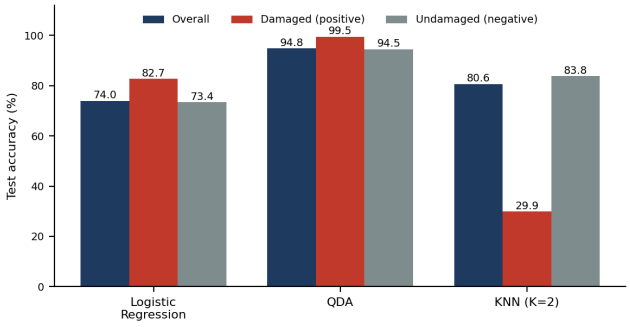
## 4. Implementation Overview

Python with NumPy, scikit-learn (`LogisticRegression`, `QuadraticDiscriminantAnalysis`, `KNeighborsClassifier`, `resample`) and Matplotlib. `util.py` loads CSVs (columns beginning with *x* are features, *y* the label); `predict_x2_x4_x5.py` is the main experiment (resample → fit → score, repeated over ratios and iterations, saving plots and accuracy tables to `Output/`); `KNN_optimal.py` selects K; `predict_x4_x5.py` repeats the study with only magnitude and distance; `plot_data.py` draws the data. The whole pipeline runs in under two minutes with `bash run_all.sh` (see README).

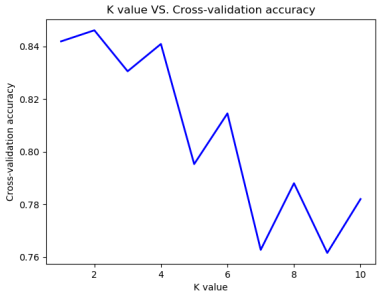
## 5. Results

| Model | Train acc. | Test acc. | Test acc. (damaged) | Test acc. (undamaged) |
|---|---|---|---|---|
| Logistic Regression | 95.3% | 74.0% | 82.7% | 73.4% |
| **QDA** | **97.8%** | **94.8%** | **99.5%** | **94.5%** |
| KNN (K = 2) | 99.1% | 80.6% | 29.9% | 83.8% |

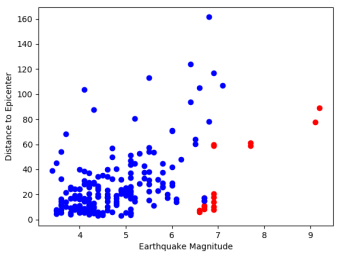
*Mean over 100 bootstrap iterations and 5 resampling ratios (values re-checked by re-running the code; they vary by a few tenths of a percent between runs).*




*Test accuracy per model and class*




*Validation accuracy vs K (best K = 2)*




*Data: magnitude vs distance (red = damaged)*

## 6. Conclusions

- **QDA is the best model** (94.8% test accuracy, about 21 points above Logistic Regression and 14 above KNN) and is stable as the resampling ratio grows (96.7% at ratio 0.2 to 93.6% at 1.0). Its Gaussian class-conditional assumption with a

separate covariance per class fits this data better than a linear boundary.

- **Logistic Regression** under-fits the non-linear class boundary: test accuracy falls from 91.5% to 65.7% as more resampled positives are added. **KNN** memorises the training set (99.1%) but detects only about 30% of damaged test bridges.
- **Limitations:** the test set has only **3 damaged bridges**, so the damaged-class accuracies are very coarse and the headline accuracy is dominated by the undamaged class. Bootstrap resampling duplicates existing positives and adds no new information, and “no recorded damage = undamaged” is an assumption. Results should therefore be read as a comparison of methods, not as a validated engineering tool.
- **Future work:** collect more damaged-bridge records, add material type, structure type and traffic features, try other models, and move to multi-class damage levels (none / moderate / collapse).